USER GUIDE: ZAT (Z_AUTH_SU53_ANALYZER)

Advanced SU53 Authorization Analyzer & Self-Service Remediation Cockpit

Transaction Code: ZAT
Version: v2.0 (Dual-Grid ALV)
Target Audience: Role Testers / Key Users / UAT Team

1. INTRODUCTION & BUSINESS PURPOSE

During SAP S/4HANA testing cycles (Unit Testing, Integration Testing, UAT), testers frequently encounter authorization failures (blocked transactions or missing business document authorizations). Conventionally, testers capture screenshot of standard transaction SU53, send it to Security Administrators, and wait for manual profile adjustments in PFCG. This causes substantial test delays and project bottlenecks.

The **Z_AUTH_SU53_ANALYZER (Transaction Code: ZAT)** utility is an automated cockpit designed to empower testers:

- **Multi-Failure SU53 History:** Aggregates all failure records from the past **3 hours** directly from kernel buffers (unlike standard SU53 which displays only the single last failed check).
- **Live Role Coverage Analysis:** Automatically cross-references missing checks against the tester's assigned Single Roles (`@08@ [OK]` or `@0A@ [ERR]`).
- **1-Click Instant Remediation:** Enables testers to select missing permissions and inject them directly into their test Single Role. The program **regenerates authorization profiles in background**, enabling immediate test resumption!

Feature Comparison: Standard SU53 vs ZAT Cockpit

Comparison Criteria	Standard SAP SU53 Transaction	Advanced ZAT Cockpit
Failure Retention Scope	Stores only the **very last authorization failure**. Executing any other action completely overwrites the buffer.	3-Hour History Extracts and aggregates all failed authorization checks over 3 hours via SUSR_USER_SU53_READ.
Assigned Role Evaluation	Displays no role information. Testers cannot identify which role is supposed to grant the missing permission.	Evaluates active Single Roles assigned to the user in real time, displaying status with intuitive visual indicators.
Remediation Capability	Read-only display. Requires manual ticket submission to Security Admin for PFCG configuration.	Integrated **[Add to Role]** action: Automatically updates role menu, populates 100% mandatory fields from TOBJ, and compiles profiles.

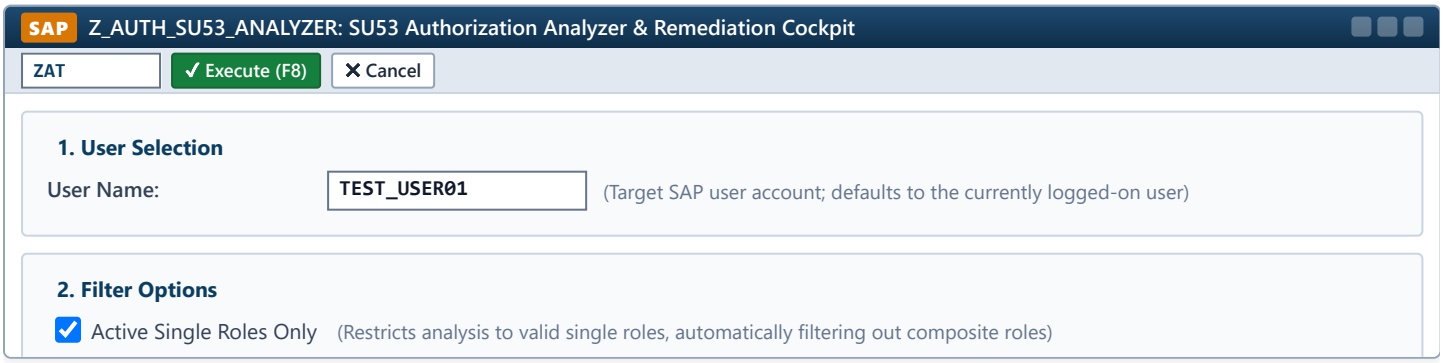
2. HOW TO LAUNCH THE PROGRAM

Testers can execute the cockpit via either of the following methods in SAP GUI:

- **Option 1 (Recommended):** Enter transaction code ZAT or `/nZAT` into the Command Field at the top-left corner and press Enter.
- **Option 2:** Open transaction SE38 or SA38, enter program name `Z_AUTH_SU53_ANALYZER`, and click **Execute (F8)**.

3. SELECTION SCREEN & PARAMETERS

The initial selection screen specifies the target user ID and active role filtering:

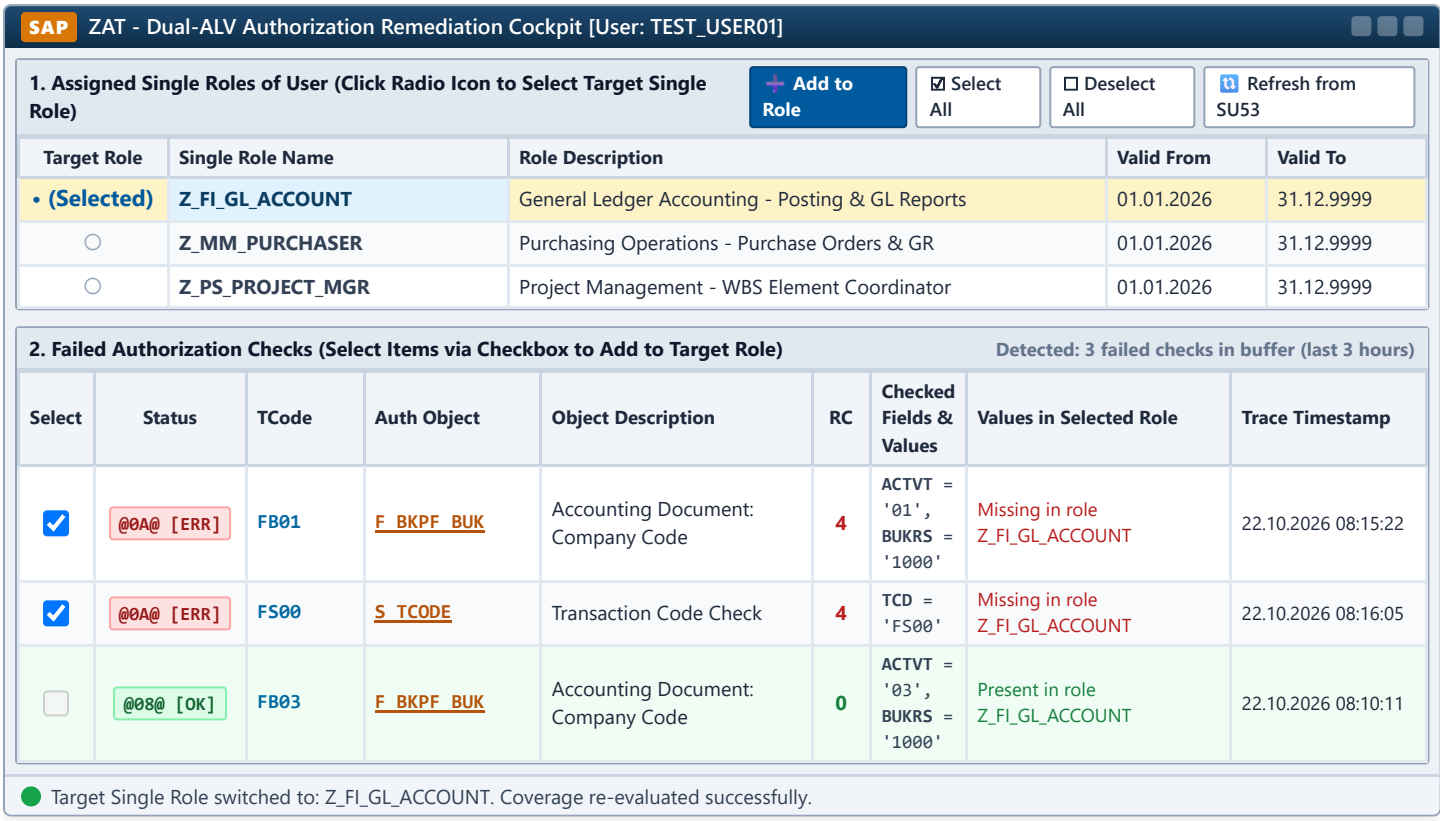


Tester Note: The user field defaults to your active account. Test Leads investigating errors on behalf of another tester can enter the colleague's SAP User ID and press **F8**.

4. DUAL-GRID COCKPIT ARCHITECTURE

Upon executing (**F8**), the cockpit opens a dual-level operational workbench:

- **Top Grid (Docking Container):** Lists all active Single Roles assigned to the user. The `Target Role` column contains interactive radio buttons allowing you to designate which role will receive authorizations.
- **Bottom Grid (Fullscreen ALV):** Displays all failed authorization checks from the past 3 hours. Status badges automatically recalculate whenever a different role is selected above.



Toolbar Actions & Navigation Controls

Action / Button	Shortcut / Icon	Functional Description
[Add to Role]	Icon ➕ (Dark Blue)	Injects all checked authorizations and TCodes into the currently selected Single Role and triggers dark profile regeneration.
[Select All]	Icon ☑	Selects all missing authorization items (`@0A@ [ERR]`). Items already satisfied (`@08@`) are automatically bypassed.
[Deselect All]	Icon ☐	Clears all selected checkboxes in the bottom grid.
[Refresh from SU53]	Icon 🔃 (or F8 key)	Re-reads system failure buffers when the tester performs additional transaction actions in another session.
Click Radio Button	`Target Role` column	Switches the active remediation role. The bottom grid immediately re-evaluates all failure rows against the newly chosen role.

5. STANDARD 5-STEP REMEDIATION WORKFLOW FOR TESTERS

Follow this standardized 5-step procedure when blocked by an authorization failure during business scenario execution:

1 Encounter authorization failure in business transaction

While executing your scenario (e.g. `FB01`, `ME21N`, `FS00`), SAP halts execution and outputs a red status bar message:
`• You are not authorized to use transaction FB01` or `• No authorization for Company Code 1000 (Object F_BKPF_BUK)`.

2 Launch ZAT in a new parallel session

Without exiting your active transaction, open a new SAP GUI window by typing `/oZAT` in the Command Field. Verify your User Name and press **F8** to retrieve failure buffer logs.

3 Select the target Single Role in Top Grid

Click the circular radio button in the `Target Role` column corresponding to the functional role you are validating (e.g. `Z_FI_GL_ACCOUNT`). The failure rows in the bottom grid instantly recalculate their coverage status.

4 Select the missing authorizations in Bottom Grid

Tick the checkboxes in the `Select` column for rows marked with red indicator `@0A@ [ERR]`. Alternatively, click the **[Select All]** toolbar button to select all valid remediation items at once.

5 Click [Add to Role] for automated remediation

Click the **[Add to Role]** button on the Top Grid toolbar. The utility automatically executes in background:

- **Role Menu Injection:** For missing transaction code checks (`S_TCODE`), injects the TCode into the role menu via standard BAPI.
- **Authorization Object Population:** Injects the missing object with 100% mandatory fields populated from the SAP standard dictionary (`TOBJ`).
- **Organizational Level Defaulting:** Automatically sets unmaintained Org Levels to * (full test authorization).
- **Dark Profile Compilation:** Recompiles the authorization profile and synchronizes the active User Buffer.

Simulated Status Message Following Remediation

● 1 transaction(s) added to menu. 1 object(s) added/updated. Profile regenerated. Data refreshed.

Outcome: As soon as the green status message appears, return to your initial transaction window (e.g. `FB01`) and press `Enter` or re-execute. The transaction will proceed smoothly without requiring a system logoff!

6. STATUS CODE REFERENCE & ADVANCED FEATURES

ALV Grid Status Indicators

Status Code	Indicator	Technical Meaning	Recommended Action
`@08@ [OK]`	• GREEN	The authorization object and checked values **already exist** in the selected role. Selection checkbox is disabled.	No action required. The failure is already satisfied by this role or granted previously.
`@0A@ [ERR]`	• RED	The authorization object or field values are **missing** in the selected role. This is the direct cause of the block.	Tick the checkbox and click **[Add to Role]** to resolve the failure immediately.
`@09@ [WARN]`	• YELLOW	Partial match; or the SU53 buffer contains no failed authorization checks within the last 3 hours.	Re-perform the failing transaction step in your business window, then click **[Refresh from SU53]**.

Advanced Operational Features

- **Authorization Object Jump (SU03):** Double-click any entry in the `Auth Object` column (e.g. `F_BKPF_BUK`). The cockpit automatically opens transaction `SU03`, displaying official SAP documentation and technical field definitions for that object.
- **Spreadsheet Export for Audit & Reporting:** Press `Ctrl + Shift + F7` or navigate to `List → Export → Spreadsheet`. Because status codes include explicit text tags (`@08@ [OK]`, `@0A@ [ERR]`), exported Excel files can be filtered cleanly by test leads or PMO teams.

7. FREQUENTLY ASKED QUESTIONS & TROUBLESHOOTING (FAQ)

1. Why does [Add to Role] display "Role is locked (Edit mode)"?
Cause: The target role is currently opened in Change Mode in transaction `PFCG` by another administrator or your own user.
Remedy: Exit edit mode in `PFCG` (enter `/n`), return to `ZAT`, and re-click [Add to Role].

2. Why is the newly added transaction not visible in the SAP Easy Access tree?
Explanation: ZAT successfully injects the TCode into the role menu and generates the technical `S_TCODE` authorization. To refresh the visual SAP Easy Access tree on the initial screen, simply type `/n` or start the transaction directly from the Command Field.

3. Are the injected authorizations permanent?
Answer: Yes. Authorizations are written directly into standard SAP tables (`AGR_1250`, `AGR_1251`, `AGR_1252`) and compiled into the generated profile. They persist permanently until the role is deliberately modified in a subsequent release cycle.

Technical Support Escalation:
For structural authorization issues (HR Org Management) or complex custom authority checks, contact the SAP Security Administration Team for specialized assistance.